

Guided Media

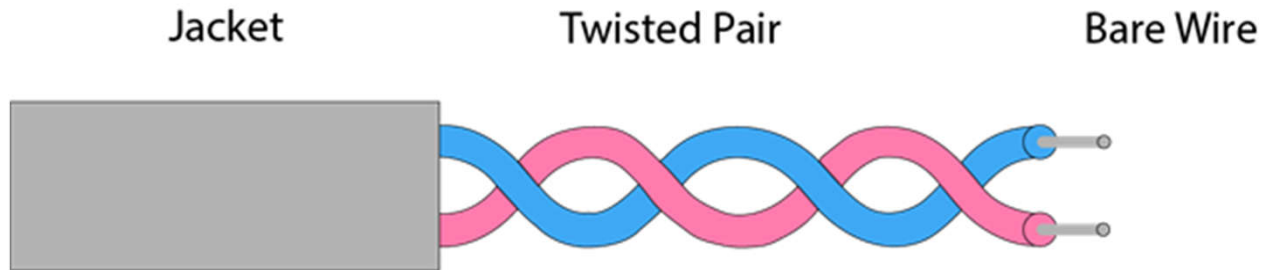
It is defined as the physical medium through which the signals are transmitted. It is also known as Bounded media.

Types Of Guided media:

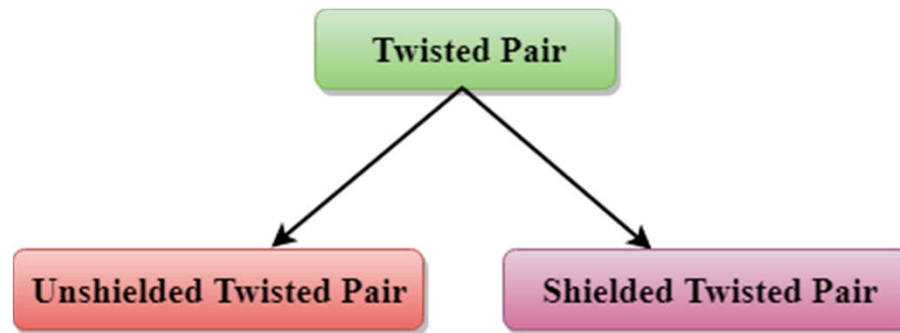
Twisted pair:

Twisted pair is a physical media made up of a pair of cables twisted with each other. A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable. **The frequency range for twisted pair cable is from 0 to 3.5KHz.**

A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern..



Types of Twisted pair:



Unshielded Twisted Pair:

An unshielded twisted pair is widely used in telecommunication. Following are the categories of the unshielded twisted pair cable:

Category 1: Category 1 is used for telephone lines that have low-speed data.

Category 2: It can support upto 4Mbps.

Category 3: It can support upto 16Mbps.

Category 4: It can support upto 20Mbps. Therefore, it can be used for long-distance communication.

Category 5: It can support upto 200Mbps.

Advantages Of Unshielded Twisted Pair:

It is cheap.

Installation of the unshielded twisted pair is easy.

It can be used for high-speed LAN.

Disadvantage:

This cable can only be used for shorter distances because of **attenuation(loss of signal)**. Signals degrade over distance in any cable, a phenomenon called attenuation, which limits the effective length for reliable signal transmission. As a signal travels through a cable, some of its energy is lost as heat, weakening the signal. **Noise Interference:**

Signals can also degrade due to external electrical currents and radio frequencies, which can be picked up by the cable.

Shielded Twisted Pair(STP)

A shielded twisted pair is a cable that contains the mesh(made from synthetic materials like polyester and nylon) surrounding the wire that allows the higher transmission rate.

Characteristics Of Shielded Twisted Pair:

The cost of the shielded twisted pair cable is not very high and not very low.

An installation of STP is easy.

It has higher capacity as compared to unshielded twisted pair cable.

It has a higher attenuation.

It is shielded that provides the higher data transmission rate.

Disadvantages



Coaxial Cable

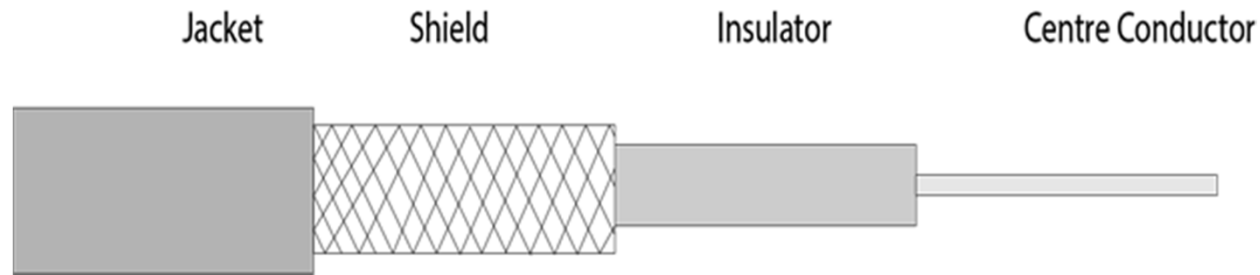
Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.

The name of the cable is coaxial as it contains two conductors parallel to each other.

It has a higher frequency as compared to Twisted pair cable.

The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.

The middle core is responsible for the data transferring whereas the copper mesh prevents from the EMI (Electromagnetic interference).



Coaxial cable is of two types:

Baseband transmission: It is defined as the process of transmitting a single signal at high speed.

Broadband transmission: It is defined as the process of transmitting multiple signals simultaneously.

Advantages Of Coaxial cable:

The data can be transmitted at high speed.

It has better shielding as compared to twisted pair cable.

It provides higher bandwidth.

Disadvantages Of Coaxial cable:

It is more expensive as compared to twisted pair cable.

If any fault occurs in the cable causes the failure in the entire network.

Fibre Optic

Fibre optic cable is a cable that uses electrical signals for communication.

Fibre optic is a cable that holds the optical fibres coated in plastic that **are used to send the data by pulses of light**.

The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring.

Fibre optics provide faster data transmission than copper wires.

Diagrammatic representation of fibre optic cable:



Basic elements of Fibre optic cable:

Core: The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre. **The more the area of the core, the more light will be transmitted into the fibre.**

Cladding: The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fibre. **The higher the refractive index the slower the light travels, which causes a correspondingly increased change in the direction of the light within the material.**

Jacket: The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fibre protection.

Following are the advantages of fibre optic cable over copper:

Greater Bandwidth: The fibre optic cable provides more **bandwidth** as compared copper. Therefore, the fibre optic carries more data as compared to copper cable.

Faster speed: Fibre optic cable carries the data in the form of light. This allows the fibre optic cable to carry the signals at a higher speed.

Longer distances: The fibre optic cable carries the data at a longer distance as compared to copper cable.

Better reliability: The fibre optic cable is more reliable than the copper cable as it is immune to any temperature changes while it can cause obstruct in the connectivity of copper cable.

Thinner and Sturdier: Fibre optic cable is thinner and lighter in weight so it can withstand more pull pressure than copper cable.

UnGuided Transmission

An unguided transmission transmits the electromagnetic waves without using any physical medium. Therefore it is also known as **wireless transmission**.

In unguided media, air is the media through which the electromagnetic energy can flow easily.

Unguided transmission is broadly classified into three categories:

Radio waves

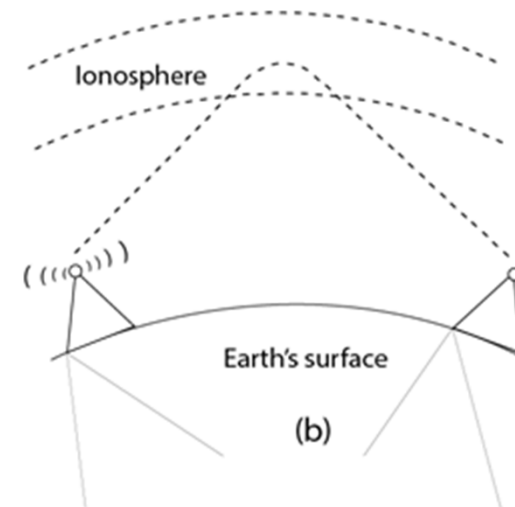
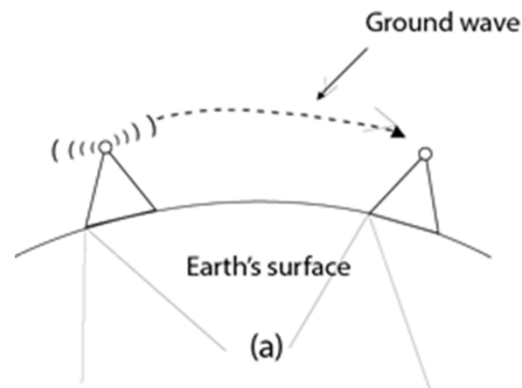
Radio waves are the electromagnetic waves that are transmitted in all the directions of free space.

Radio waves are **omnidirectional**, i.e., the signals are propagated in all the directions.

The range in frequencies of radio waves is from **3Khz to 1 khz**.

In the case of radio waves, **the sending and receiving antenna are not aligned, i.e., the wave sent by the sending antenna can be received by any receiving antenna.**

An example of the radio wave is **FM radio**.



Applications Of Radio waves:

A Radio wave is useful for multicasting when there is one sender and many receivers. An FM radio, television, cordless phones(landline) are examples of a radio wave.

Advantages Of Radio transmission:

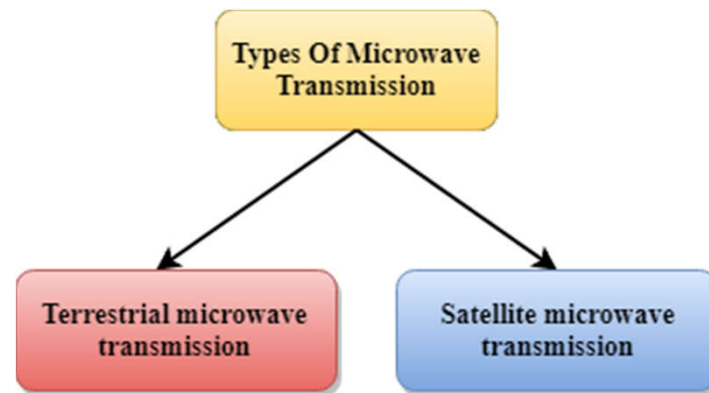
Radio transmission is mainly used for wide area networks and mobile cellular phones.

Radio waves cover a large area, and they can penetrate the walls.

Radio transmission provides a higher transmission rate.

Microwaves are of two types:

- Terrestrial microwave
- Satellite microwave communication.



Terrestrial Microwave Transmission

Terrestrial Microwave transmission is a technology that transmits the focused beam of a radio signal from one ground-based microwave transmission antenna to another.

Microwaves are the electromagnetic waves having the frequency in the range from 1GHz to 1000 GHz.

Microwaves are unidirectional as the sending and receiving antenna is to be aligned, i.e., the waves sent by the sending antenna are narrowly focussed.

In this case, antennas are mounted on the towers to send a beam to another antenna which is km away.

It works on the line of sight transmission, i.e., the antennas mounted on the towers are the direct sight of each other.

Microwave transmission is the process of sending information, such as voice, data, and TV signals, using electromagnetic waves in the microwave frequency range (300 MHz to 300 GHz). These high-frequency waves allow for large amounts of data to be transmitted at high speeds over long distances, making them ideal for point-to-point and satellite communication systems.



Characteristics of Microwave:

Frequency range: The frequency range of terrestrial microwave is from 4-6 GHz to 21-23 GHz.

Bandwidth: It supports the bandwidth **from 1 to 10 Mbps**.

Short distance: It is inexpensive for short distance.

Long distance: It is expensive as it requires a higher tower for a longer distance.

Attenuation: Attenuation means loss of signal. It is affected by environmental conditions and antenna size.

Advantages Of Microwave:

Microwave transmission is cheaper than using cables.

It is free from land acquisition as it does not require any land for the installation of cables.

Microwave transmission provides an easy communication in terrains as the installation of cable in terrain is quite a difficult task.

Communication over oceans can be achieved by using microwave transmission.

Disadvantages of Microwave transmission:

Eavesdropping: An eavesdropping creates insecure communication. Any malicious user can catch the signal in the air by using its own antenna.

Out of phase signal: A signal can be moved out of phase by using microwave transmission.

Susceptible to weather condition: A microwave transmission is susceptible to weather condition. This means that any environmental change such as rain, wind can distort the signal.

Bandwidth limited: Allocation of bandwidth is limited in the case of microwave transmission.

Satellite Microwave Communication

A satellite is a physical object that revolves around the earth at a known height.

Satellite communication is more reliable nowadays as it offers more flexibility than cable and fibre optic systems.

We can communicate with any point on the globe by using satellite communication.

How Does Satellite work?

The satellite accepts the signal that is transmitted from the earth station, and it amplifies the signal. The amplified signal is retransmitted to another earth station.

Advantages Of Satellite Microwave Communication:

- The coverage area of a satellite microwave is more than the terrestrial microwave.
- The transmission cost of the satellite is independent of the distance from the centre of the coverage area.
- Satellite communication is used in mobile and wireless communication applications.
- It is easy to install.
- It is used in a wide variety of applications such as weather forecasting, radio/TV signal broadcasting, mobile communication, etc.

Disadvantages Of Satellite Microwave Communication:

- Satellite designing and development requires more time and higher cost.
- The Satellite needs to be monitored and controlled on regular periods so that it remains in orbit.
- The life of the satellite is about 12-15 years. Due to this reason, another launch of the satellite has to be planned before it becomes non-functional.

Infrared

An infrared transmission is a wireless technology used for communication over short ranges.

The frequency of the infrared is in the range from 300 GHz to 400 THz.

It is used for short-range communication such as data transfer between two cell phones, TV remote operation, data transfer between a computer and cell phone resides in the same closed area.

Characteristics Of Infrared:

It supports high bandwidth, and hence the data rate will be very high.

Infrared waves cannot penetrate the walls. Therefore, the infrared communication in one room cannot be interrupted by the nearby rooms.

An infrared communication provides better security with minimum interference.

Infrared communication is unreliable outside the building because the sun rays will interfere with the infrared waves.